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Multi-Task Graph Neural Network for Truck Speed Prediction Under Extreme Weather Conditions

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ABSTRACT

Truck speed prediction plays a key role in truck transportation management. However, it is a very challenging task since the truck traffic usually shows complex patterns. Most of the existing traffic prediction methods lack the ability to model the dynamic spatial-temporal correlations of truck traffic or ignore contributing contextual factors that impact traffic. Also, truck traffic data is typically sparse and noisy, which makes the truck speed prediction an even more challenging task. How to improve the truck speed prediction by taking advantage of other relevant truck traffic information (such as the truck flow) has not been investigated in depth. Additionally, traffic congestions and poor driving conditions caused by extreme weather conditions can make sudden changes in the general pattern of the truck speed. In this paper, we propose a novel Multi-Task Context Based Gated Recurrent Unit Graph Convolutional Network (MT-C2G) to predict the truck speed under extreme weather conditions. MT-C2G includes four major components: The spatial dependence learning component captures the spatial dependencies shaped by the topological structure of the road network. Truck traffic feature temporal dependence modeling component is built to acquire the temporal dependencies involved in the truck traffic features, and contextual feature temporal dependence modeling component employs a layer of GRU units to capture the temporal dependencies of contextual factors. The multi-task learning component then leverages the information between the truck speed and flow prediction tasks through attention mechanism for improving the performance. Moreover, a data augmentation method SMOTE is utilized to balance the data with the extreme weather conditions. Experiments on two real datasets demonstrate that the proposed MT-C2G fairly outperforms six state-of-the-art traffic prediction methods.

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CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Information systems** → **Spatial-temporal systems**; • **Applied computing** → **Transportation**.

KEYWORDS

truck speed prediction, multi task learning, spatiotemporal models, extreme weather, truck traffic forecasting

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1 INTRODUCTION

Reliable truck freight transport ensures the sustainable operation of modern supply chains and the resulting economic development. Truck speed prediction plays a key role in truck transportation management. To predict the truck speed, one straightforward solution is to use the passenger car data and existing traffic prediction models. However, there are distinct characteristics between the speed patterns of trucks and passenger cars. The passenger car speed, for example, shows clear patterns of morning and evening peaks, and the car volume in rush hours are usually busier than at night, while truck traffic does not necessarily follow these patterns. Moreover, most countries have special operational policies to restrict truck operation on specific road sections at specific times of the day. As a result, nights in those areas are often busier for trucks than days. Additionally, even the best plans can be disrupted by extreme weather conditions, especially when it causes adverse driving condition and road closures. Snowstorm, ice, fog and floods can cause truck driving conditions to be extremely difficult and dangerous. Over the past few years, there has been a rapid increase in the number of trucks equipped with GPS devices. As a result, a large volume of truck trajectory data are generated, which contains information about the locations of trucks as well as some basic information such as speed and heading. It is possible to develop highly effective traffic information services based on such a large amount of truck trajectory data, which can assist in analyzing traffic operation status for trucks.

In light of all of the above, the following questions with regards to truck speed prediction under extreme weather conditions have not been thoroughly investigated:

- (1) *How can spatio-temporal and other contextual information be incorporated with historical truck trajectories to achieve the best truck speed prediction results?* Although there is a wide variety of research on traffic speed forecasting, most existing efforts have only focused on passenger car traffic prediction. A few studies exist to use the truck GPS trajectory data with applying deep learning models for the truck speed prediction. Zhao et al. [20] developed an enhanced GRU to forecast the truck speed on urban express highways under non-recurrent overcrowded conditions by using the truck GPS data. Wang et al. introduced a GPS data expansion and prediction methods based on LSTM and GRU method [15]. Despite focusing on the truck speed prediction, these studies only consider the temporal dependence of the truck speed and do not fully exploit the topological spatial features of the road network. Additionally, accurate truck speed prediction is not only depends on the historical truck traffic information but also other contextual features, such as weather condition, holiday or non-holiday days and special events influence the speed variation in road networks. As a result, capturing the correlation between truck speed and contextual features will provide a better understanding of forecasting truck speed patterns.
- (2) *How can improve truck speed prediction accuracy by taking advantage of other relevant truck traffic information (such as the truck flow)?* Training supervised deep learning models requires a large amount of labeled data. However, truck GPS samples are often contaminated with noise and sparsity, resulting in insufficient data to train an accurate learner during the training process. This will negatively affect deep learning models' accuracy. It has been shown that correlations between the speed and flow have a clear pattern, and capturing this pattern can not be neglected [18]. Consequently, by utilizing the relatedness between these two historical truck traffic variables (i.e., speed and flow), it is possible to improve the learning ability of the prediction task.
- (3) *How to deal with under-represented truck traffic data under extreme weather conditions?* One of the other challenges for truck speed prediction under extreme weather conditions is that the machine learning algorithms detect dominant patterns in the data and neglect capturing under-represented samples of data, such as extreme weather conditions. The extreme weather conditions are less likely to happen throughout the year, but they are notably important for the truck speed prediction.

This paper proposes a novel Multi-Task Context Based Gated Recurrent Unit Graph Convolutional Network (MT-C2G) for the truck speed prediction under the extreme weather conditions. To the best of our knowledge, this is the first study to apply the Multi-Task Learning (MTL) to describe the actual correlation between the truck speed and flow to predict the truck speed, along with using the oversampling technique on truck GPS data under the extreme

weather conditions. Specifically, our contributions are listed as follows:

- MT-C2G takes the contextual features into account when predicting truck speed. It employs a layer of GRU units to capture the temporal dependence in multivariate time series of external factors, including temporal patterns shown in the historical truck traffic and contextual factors. Since weather is a time-varying factor that directly affects the road condition and traffic state, it is crucial to capture the temporal dependencies of this factor in speed prediction task.
- MT-C2G utilizes MTL model to obtain a reasonable speed prediction based on real-world truck speed and flow historical data. Considering truck speed and flow simultaneously, MT-C2G makes full use of the interactions between these two variables, as well as reducing the impact of noise and sparseness in the data, focusing on the relevant feature that provides more evidence, reducing the likelihood of over-fitting, and learning a more general representation of the data.
- Truck speed and flow samples that are synchronized with extreme weather conditions are under-represented, but the samples should be emphasized during the training. To address the issue, MT-C2G utilizes the Synthetic Minority Oversampling Technique (SMOTE) to over-sample the data under the extreme weather conditions, which improves the truck speed prediction accuracy for extreme weather conditions.
- We conduct extensive experiments on two real-world datasets, i.e. Calgary and Brussels truck traffic data, for evaluating the performance of MT-C2G model. Results prove that the proposed approach obtains significant improvement over state-of-the-art baseline methods.

The rest of the paper is organized as follows: Section 2 provides an overview of the previous studies and sheds light on the challenges of truck speed prediction. Afterwards, Section 3 covers the technical details of the proposed MT-C2G method. In Section 4, experiments are conducted to compare MT-C2G with baselines. Finally, Section 5 discusses the conclusions and future works.

2 RELATED WORKS

In this section, we first review some of the literature on using deep learning models for the truck traffic prediction, and then discuss methods for passenger car traffic prediction that have shown successful results recently.

In recent years, few studies have investigated the truck traffic speed in detail, especially using GPS data. For instance, using the GPS data generated by the trucks, Zhao et al. [20] developed an enhanced GRU algorithm to forecast the truck traffic speed on urban express highways under non-recurrent overcrowded conditions. In order to match GPS samples with their corresponding road segments, they first employ a GPS map matching algorithm. Afterwards, the trucks' travel data is extracted from the original data. In another study, Wang et al. introduced a GPS data expansion and prediction methods based on LSTM and GRU method [15]. First they used a piece-wise constant coefficient method to expand the

sampled truck traffic flow and then they used LSTM and GRU to perform the actual truck traffic flow predictions task. Although these two studies on truck traffic flow and speed prediction improved prediction accuracy, they only consider the temporal dependency of traffic status and fail to take full advantage of the spatial features.

In terms of passenger vehicle traffic prediction, the existing models can be categorized into model-driven approaches and data-driven approaches. Model-driven approaches such as the queuing theory model [17], cell transmission model [16], and traffic velocity model [14], often require a complete understanding of the system environment based on prior knowledge. In reality, however, traffic data is affected by various factors, making it hard for model-driven models to accurately describe the variations of traffic data in complex real-world environments[21].

By contrast, data-driven approaches automatically determine the regularity of input data. Traditional statistical models such as Autoregressive Integrated Moving Average (ARIMA) and machine learning models such as Support Vector Regression (SVR), K-Nearest Neighbor(KNN) are two significant representatives of data-driven methods. In recent years, however, deep learning methods have been widely applied to traffic prediction problems with superior results than traditional statistical and machine learning models. Some of the deep learning models including the FNN [11], LSTM [22], RNN [13], GRU [5], only take the temporal dependency into account and ignore the spatial correlations of the road network. However, recent studies have shown that spatial correlations also contribute significantly to traffic prediction accuracy [7]. Therefore, researchers have extended deep learning approaches to graph data to exploit network structure information and proposed graph convolutional neural networks (GCNs) that are specifically designed for processing network structures [7, 23]. These models have demonstrated significant improvements in capturing spatial dependence between road segments, resulting in better performance on traffic prediction tasks [8, 21]. As an instance, Zhao et al. [21] in their study proposed a neural network based approach for traffic forecasting called T-GCN, which combines the GCN to capture the topological structure of the graph to obtain the spatial dependence and the GRU model to capture the dynamic change of node attribute to obtain the temporal dependence. However, in addition to historical traffic information, variety of contextual factors such as weather conditions may effect the traffic states, and above mentioned methods fail to consider such factors. To enhance the spatiotemporal accuracy in predicting traffic, Zhu et al. [24] proposed a method called AST-GCN to model the contextual factors (i.e., weather conditions and surrounding POI distribution). AST-GCN models the weather information as dynamic attributes and surrounding POIs as static attributes and design an attribute-augmented unit to encode and integrate those factors into the spatiotemporal graph convolution model. Their study verifies the importance of considering contextual information in traffic forecasting tasks.

In all the models discussed up to this point, learning happens within the context of single-task learning (STL), which fails to make use of information sharing among related tasks. In contrast to STL, MTL has the potential to improve the prediction performance by transfer information between related tasks during the training. Several attempts were made to perform MTL to explore spatiotemporal

patterns and capture nonlinear mapping correlations in traffic prediction. Zhang et al.[19] present a deep learning based multi-task model to capture the correlating between various links, and to predict traffic speeds of multiple road links simultaneously. In another attempt, Zhang et al. [18] proposed a novel MTL framework with GRU and residual mappings to improve the forecasting accuracy of traffic flow and speed simultaneously. Also, this study utilizes the statistical tools to conduct feature engineering to extract the most informative features for the proposed MTL-GRU model. Buroni et al. [1] also proposed AST-MTL to predict the traffic flow and speed simultaneously at the road network scale. AST-MTL aggregates a multilayer fully-connected neural network (FNN) and a multi-head attention mechanism to learn related tasks while improving generalization performance. The model also uses the GCN to capture the topological structure of the road network and utilizes the GRU to model the temporal features of traffic condition. Experimental results of AST-MTL proves the effectiveness of multi-task learning in performing multi-horizon traffic prediction task, yet modeling the effects of contextual factors is neglected in this study.

These models are developed to capture the correlation between traffic flow and speed of passenger vehicles. However, as mentioned before, one of the issues that have not been considered in mentioned methods is the effects of contextual factors in truck speed prediction task which may negatively influence the prediction accuracy. Added to that, due to the lack of sufficient data when dealing with truck speed prediction under the extreme weather conditions, it is important to make sure the model is able to accurately model the patterns for under-represented samples (i.e. traffic speed during extreme weather conditions) as well.

3 METHODOLOGY

3.1 Problem Statement

In this section, we introduce the definitions, and state the problem that we intend to solve.

In a transportation road network, each road can be subdivided into multiple segments according to the intersections on the road. As our goal is to predict the truck speed on the road segment, an interchanged road network is introduced in this paper to describe the topological structure of the road network.

Definition 1 (Interchanged Road Network): An interchanged road network is an unweighted graph denoted by $G = (V, E)$ where $V = \{v_1, v_2, \dots, v_R\}$ corresponds to the set of vertices in G , representing the road segments of the road network, and $E = \{e_1, e_2, \dots, e_I\}$ represents the set of edges contained in G . Two vertices in graph G are adjacent if and only if the corresponding road segments are connected in the road network. The adjacency matrix of the interchange road network G is denoted by $A_G \in \mathbb{Z}^{|V| \times |V|}$. The value of each element $a_{i,j}$, where $i, j \in V$, is 1 if and only if there exists an edge $e_k \in E$ between those two vertices, and 0 if otherwise.

To represent the historical truck traffic information on the road segment, truck traffic feature tensor is defined for vertices of the interchanged road network.

Definition 2 (Truck Traffic Feature Tensor): Given the set of vertices $V = \{v_1, v_2, \dots, v_R\}$ of the interchanged road network G and a set of time slices $T = \{t_1, t_2, \dots, t_N\}$, a truck traffic tensor is expressed as $X \in \mathbb{R}^{|V| \times |T| \times 2}$, where $|V|$ stands for the cardinality of

the graph G 's set of vertices and $|T|$ is the length of time slices of the historical truck traffic information. The elements of the tensor are denoted by $x_{v_r}^{t_s, \Gamma}$, where $\Gamma \in \{Speed, Flow\}$. $x_{v_r}^{t_s, Speed}$ corresponds to the average speed of the trucks on the vertex v_r at time slice t_s . Similarly, $x_{v_r}^{t_s, Flow}$ is the trucks' traffic flow for the vertex v_r at time slice t_s .

Definition 3 (Contextual Feature Tensor): Given the set of vertices $V = \{v_1, v_2, \dots, v_R\}$ of the interchanged road network G and a set of time slices $T = \{t_1, t_2, \dots, t_N\}$, a contextual feature tensor, denoted by C , is a tensor with dimension $|V| \times |T| \times p$, where p stands for the number of contextual features. Contextual features describe the contributing factors other than the historical truck traffic information that affect the truck traffic. The contextual features considered in this paper includes:

- **Weather:** Different weather conditions are classified and assigned an integer value to each class.
- **Business day/Holiday:** The value corresponding to this feature takes value 1 if the day is a business day, and 0 if it is a holiday.
- **Day of the week:** A discrete value is used to represent each day of the week.
- **Time of the day:** This feature shows different time of the day, such as in hours.

Thus, a joint truck speed and flow prediction problem can be defined as learning the function f from an interchanged road network G , truck traffic feature tensor X , and corresponding contextual feature tensor C to predict truck speed and flow for the future τ consecutive time slices.

$$[\tilde{x}^{t_{s'}, Speed}, \tilde{x}^{t_{s'}, Flow}] = f([X|C]; G) \quad (1)$$

where $\tilde{x}^{t_{s'}, Speed}$ and $\tilde{x}^{t_{s'}, Flow}$ are standing for predicted truck speed and flow at time slice $t_{s'} \in \{t_{N+1}, \dots, t_{N+\tau}\}$.

For each task $\Gamma \in \{Speed, Flow\}$ the loss function is defined as follows:

$$\mathcal{L}^\Gamma(\theta) = \|x_{v_r}^{t_{s'}, \Gamma} - \tilde{x}_{v_r}^{t_{s'}, \Gamma}\| + \lambda^\Gamma \mathcal{L}_{Reg}^\Gamma \quad (2)$$

for all $v_r \in V$, where $x_{v_r}^{t_{s'}, \Gamma}$ and $\tilde{x}_{v_r}^{t_{s'}, \Gamma}$ correspond to the real value of the target Γ and the predicted value of that target at time $t_{s'}$. $\mathcal{L}_{Reg}^\Gamma$ denotes the L_2 regularization term used to avoid over-fitting for task Γ , and λ^Γ is the corresponding hyperparameter. $\theta = \{\theta^{Shared} \cup \theta^{Speed} \cup \theta^{Flow}\}$ represents the set of the all trainable parameters. While θ^{Shared} denotes the learnable parameters of the shared representation, θ^{Speed} and θ^{Flow} are the parameters exclusive to each task, namely, truck speed prediction and truck flow prediction.

The objective function for the joint truck speed and flow prediction problem is formulated as:

$$\min_{\theta} w^{Speed} \cdot \mathcal{L}^{Speed}(\theta) + w^{Flow} \cdot \mathcal{L}^{Flow}(\theta) \quad (3)$$

w^{Speed} and w^{Flow} are the task specific weights for speed prediction and flow prediction tasks, respectively. In this paper, we attribute the same weight of value 1 to each task; hence, $w^{\Gamma \in \{Flow, Speed\}} = 1$.

3.2 MT-C2G's Framework

Figure 1 illustrates the overall architecture of the proposed framework. The first step in the data preprocessing is to locate the GPS track points on corresponding road segments of the road network, a task that is via a map matching technique. Then, the average truck speed and flow of each road segment are calculated and represented as time series. Next, an oversampling technique, named SMOTE [2], is utilized on the training data to improve the feature availability for each weather class and make the training data more general. The proposed MT-C2G, consisting of four major components.

- **Spatial dependence modeling:** In spatial dependence modeling component, two layers of GCN model are used to capture the spatial dependencies shaped by the topological structure of the road network.
- **Truck traffic feature temporal dependence modeling:** For truck traffic feature temporal dependence modeling, a layer consisting of a sequence of GRU units is employed to extract the temporal dependencies from the historical traffic data.
- **Multi-task learning:** With the use of attention blocks, multi-task learning component makes use of the shared information contained in the two prediction tasks, truck speed and flow prediction tasks, to improve the performance of each individual task. A dense unit is used to present a learnable shared representation of these two tasks as the query of the attention mechanism.
- **Contextual feature temporal dependence modeling:** Contextual feature temporal dependence modeling component employs a layer of GRU units are used to capture the temporal dependencies in multivariate time series of contextual factors. The novelty of MT-C2G lies in modeling the latent and residual effect of weather conditions on traffic parameters using this RNN-based model for improving the prediction performance of the model.

In the following, we will discuss each component in details.

3.3 Data Preprocessing

For each truck, the raw GPS dataset includes the truck ID, longitude, latitude, timestamp, current driving speed, and direction. To convert the raw GPS signals into the truck traffic speed and flow on the road segments, a Hidden Markov Model (HMM) [10] based algorithm for map matching is first used to match the GPS points to corresponding road segments. Then, the truck speed and flow are calculated based on the average speed and the number of GPS points on each road segment. The average speed of trucks on the road segment v_r, S_{v_r, t_s} , is calculated as follows:

$$S_{v_r, t_s} = \frac{1}{M_k} \sum_{i=1}^{M_k} \left(\frac{1}{Traj_i - 1} \sum_{j=1}^{Traj_i - 1} speed(o_{i,j}) \right) \quad (4)$$

where M_k denotes the number of trajectories passing road segment v_r , $Traj_i$ denotes the number of GPS points in the part of the trajectory i passing through the road segment v_r . $o_{i,j}$ denotes the j -th original GPS point of the trajectory i , passing through the road segment v_r . $speed(o_{i,j})$ represents the speed information of

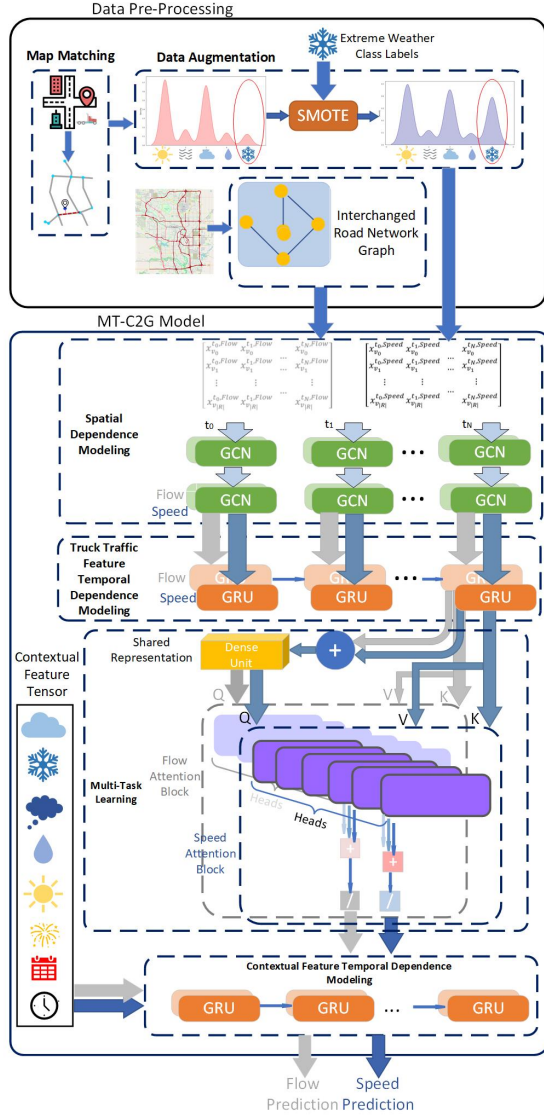


Figure 1: MT-C2G's Framework

the matching point $o_{i,j}$. The calculation results are then used to generate the truck traffic tensor.

As mentioned, when considering the impact of weather in the truck speed and flow prediction, one of the major challenges is that the frequency of days with extreme weather conditions, such as snowstorm days, are expected to be significantly less than those with normal weather conditions, such as cloudy or sunny days, which causes the number of the instances in the training dataset with those extreme weather conditions are consequently be much less frequent than those with normal weather conditions. Building a learning model with an imbalanced dataset will cause the underrepresented extreme weather classes been overlooked or even ignored. To overcome this issue, an oversampling technique, called

SMOTE [2], is utilized to improve the feature availability for each class and makes the samples more general.

3.4 Spatial Dependence Learning

One of the key steps in predicting the truck traffic targets is obtaining the local spatial dependence between the vertices on the interchanged road network G . In MT-C2G, two layers of the first-order approximation of GCN presented in [7] are used to capture spatial dependencies. The modeling process can be expressed through the following rule for each layer l :

$$Z_{l+1}^{\Gamma} = \text{ReLU} \left(\hat{D}_G^{-\frac{1}{2}} \hat{A}_G \hat{D}_G^{-\frac{1}{2}} Z_l^{\Gamma} W_l^{\Gamma} \right) \quad (5)$$

Here, ReLU represents the ReLU activation function. $\hat{A}_G$ corresponds to the adjacency matrix of graph G plus the self-connections. Hence, $\hat{A}_G = A_G + I$, where I is the identity matrix. $\hat{D}_G$ denotes the diagonal degree matrix, calculated using $\hat{D}_{i,i} = \sum_j \hat{A}_{i,j}$. W_l^{Γ} is the learnable weight matrix for the prediction task Γ . Given an interchanged road network graph $G = (V, E)$, we construct the spatial dependence learning model's initial input feature matrix $Z_0^{\Gamma} \in \mathbb{R}^{|V| \times |T|}$, where $|V|$ stand for the number of vertices in traffic feature graph and $|T|$ denotes the cardinality of the time slices' set. While Z_0^{Speed} is the frontal slice of truck traffic feature tensor X corresponding to trucks' speed, Z_0^{Flow} is that of truck traffic feature tensor, belonging to trucks' traffic flow features.

3.5 Temporal Dependence Learning

The temporal dependencies in MT-C2G model are obtained through two sets of GRU units. These two layers are explained as follows:

3.5.1 Truck Traffic Feature Temporal Dependence Modeling: Apart from capturing the spatial dependencies in road networks for truck speed and flow prediction, it is required to acquire the temporal dependencies involved in the truck speed and flow data time series. In our proposed model, we use GRU to capture the temporal dependencies in the network's truck traffic status.

3.5.2 Contextual Feature Temporal Dependence Modeling: Truck traffic speed and flow are influenced by factors other than topological structure of the road network. Weather conditions are one of the factors that exert influence on the truck traffic condition. For example, during the heavy snow or rain the speed of the vehicles experience decline because of the decreased visibility. In this study, we consider the weather condition as one of influential factors in truck speed and flow patterns. As an illustration of the influence of weather conditions in truck speed pattern, Figure 2 shows the changes in truck speeds on a road segment during different weather conditions over time on different consecutive days. It is perceivable through this figure that with the increase in the amount of snow and/or decrease in the weather temperature, the average speed of trucks should experience considerable fall. Though Figure 2, it is also observable that the impact of precipitation and extreme weather conditions in truck traffic state would not cease upon the end of a weather condition as such and residual effect will remain for consecutive period of time. To acquire the temporal

dependencies in this multivariate time series, a layer of GRU units is used.

Other contextual factors used in this paper are explained in section 3.1. The calculation process of the above-mentioned GRU parameters for each task Γ at time t_s is shown in Equation 6.

$$u^{t_s, \Gamma} = \sigma \left(W_u^\Gamma \left[F^{t_s, \Gamma}, h^{t_{s-1}, \Gamma} \right] + b_u^\Gamma \right) \quad (6a)$$

$$r^{t_s, \Gamma} = \sigma \left(W_r^\Gamma \left[F^{t_s, \Gamma}, h^{t_{s-1}, \Gamma} \right] + b_r^\Gamma \right) \quad (6b)$$

$$c^{t_s, \Gamma} = \tanh \left(W_c^\Gamma \left[F^{t_s, \Gamma}, h^{t_{s-1}, \Gamma} \right] + b_c^\Gamma \right) \quad (6c)$$

$$h^{t_s, \Gamma} = u^{t_s, \Gamma} * h^{t_{s-1}, \Gamma} + (1 - u^{t_s, \Gamma}) * c^{t_s, \Gamma} \quad (6d)$$

where $u^{t_s, \Gamma}$, $r^{t_s, \Gamma}$, and $c^{t_s, \Gamma}$ are respectively the update gate, reset gate, and memory cell value of the current time step. $h^{t_s, \Gamma}$ is the hidden state. W and b correspond to weights and biases in the training process. For the truck traffic features' temporal dependence modeling, $F^{t_s, \Gamma}$ stands for the outputs of the GCN layers at time t_s . The value correspond to $F^{t_s, \Gamma}$ for contextual features' temporal dependence modeling are the result of the corresponding attention-based model concatenated with the contextual feature tensor C .

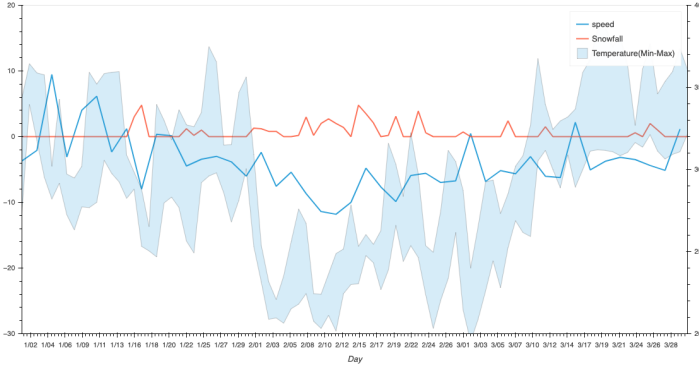


Figure 2: Illustration of the influence of weather conditions on truck speed pattern. Red shows the amount of snow, blue line shows the average speed of trucks across the city, and blue area represents the minimum and maximum weather temperature.

3.6 Multi-Task Learning Model

To improve accuracy of the predictions, instead of using a STL approach, MT-C2G takes advantage of the commonalities between the truck speed and flow prediction tasks, through jointly optimizing a shared model. One of the major advantages of information sharing as a result of this simultaneous optimization of the shared model is improving the prediction accuracy, by extracting and utilizing the shared information.

Sampled GPS data, due to the noise and sparseness, suffers from inaccuracies[15]. MTL by sharing information between the tasks, allows to mitigate the impact of noisy training data on forecasting. To take advantage of the correlation between speed and flow prediction, attention mechanism is used. Basically, by assigning weights,

the model focuses on the part of the task-specific inputs that are relevant for the prediction purpose and filters out the irrelevant ones. This allows the MTL algorithm to perform better in those tasks using the commonalities among the prediction targets rather than learning each task separately. The attention mechanism can be described as follows for each task Γ :

$$Q = [M^{Speed} | M^{Flow}] \quad (7)$$

$$K^\Gamma = V^\Gamma = M^\Gamma \quad (8)$$

$$\text{Attention}(Q, K^\Gamma, V^\Gamma) = \text{Softmax} \left(\frac{QK^\Gamma}{\sqrt{d_{att}}} \right) V^\Gamma \quad (9)$$

where d corresponds to the embedding dimension. To take advantage of parallelism, the Query, Key, and Value are mapped to H different sub-spaces. M^Γ represents the output of the first layer of GRU units.

$$[Q_1, \dots, Q_H] = [QW_{Q_1}^\Gamma, \dots, QW_{Q_H}^\Gamma] \quad (10a)$$

$$[K_1^\Gamma, \dots, K_H^\Gamma] = [K^\Gamma W_{K_1}^\Gamma, \dots, K^\Gamma W_{K_H}^\Gamma] \quad (10b)$$

$$[V_1^\Gamma, \dots, V_H^\Gamma] = [V^\Gamma W_{V_1}^\Gamma, \dots, V^\Gamma W_{V_H}^\Gamma] \quad (10c)$$

where $W_{Q_i}^\Gamma$, $W_{K_i}^\Gamma$, and $W_{V_i}^\Gamma$ are the conversion matrices.

4 EXPERIMENTS

In this section, we present the experimental results on two real-world datasets, i.e. Calgary, Canada and Brussels, Belgium, to evaluate the proposed MT-C2G.

4.1 Data Source and Data Preprocessing

The Calgary truck data is provided by American Transportation Research Institute (ATRI) for the year 2019. More than 11,000 trucks are included during the period and the total number of GPS records is nearly 22,000,000. The Bruxelles truck data, is collected from a public available dataset¹. It contains 2 months (01/01/19 -01/03/19) of HGVs' traffic in Brussels and the time resolution is 30 minutes. The road networks of Calgary and Brussels are retrieved from Open Street Map, as shown in Figure 3, and the weather data is collected from public weather APIs available for both cities.

For the data preprocessing, the Hidden Markov Model (HMM) map matching [10] method is first applied on the truck GPS datasets. After map matching, we are able to obtain GPS points information on each road segment for trucks over each time step for both datasets as explained in Section 3.3. Then we re-sample and calculate the number of passing trucks on each road segment, as well as their average speed. The road segments without speed and flow data are removed.

4.2 Experimental Settings

All experiments are conducted on Google Colab equipped with Intel(R) Xeon(R) CPU @ 2.20GHz on two Cores, NVIDIA TESLA T4 GPU, and 13GB Memory. We partition 70% of the data and perform SMOTE data augmentation to generate the training set. Figure 4 show the density features for different weather classes before and after applying the SMOTE on the city of Calgary dataset. The rest of data (30% of the original data) is used as testing set. The prediction

¹<https://www.kaggle.com/datasets/giobbu/belgium-obu>

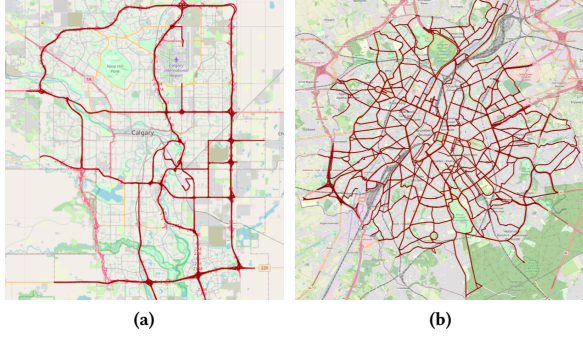


Figure 3: (a) Calgary road network, (b) Brussels road network

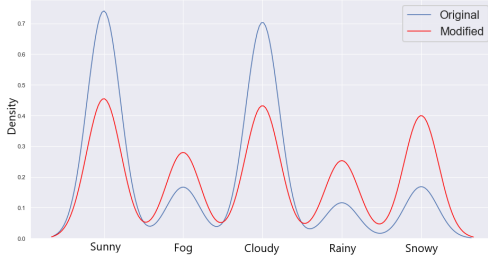


Figure 4: Comparison of density of different weather classes before and after applying SMOTE - Calgary

horizon is set to 6 on this experiment. Adam optimizer with the Learning Rate of 0.001 is used to optimize the objective function, while the batch size the value is set to 32. Finally, the value for the drop out rate is 0.1 for the experiment.

In order to assess the quality of the traffic predictions, it is essential to establish metrics that allow the comparison of the different methods. This evaluation must consist of a comparison between the predicted truck speed and the actual truck speed conditions at the selected date and time. Two metrics are used:

- Mean Absolute Error (MAE): This metric corresponds to the average absolute difference between the predicted $\hat{x}$ and the true values x [3].

$$MAE = \sum_{i=1}^n \frac{|x_t - \hat{x}_t|}{n} \quad (11)$$

- Root Mean Square Error (RMSE): This metric corresponds to the square root of the mean of the squared difference between the observed x and the predicted values $\hat{x}$.

$$RMSE = \left[\frac{1}{n} \sum_{t=1}^n (x_t - \hat{x}_t)^2 \right]^{\frac{1}{2}} \quad (12)$$

4.3 Baselines

We compare the effectiveness of the proposed MT-C2G model with the following baselines:

- History Average model (HA)[9], which uses the average truck traffic information in the historical periods as the prediction.
- Autoregressive Integral Moving Average Model (ARIMA)[6].
- Support Vector Regression (SVR)[12].
- Single Task Learning Gated Recurrent Unit model (STL-GRU) [4], which utilizes gating mechanism in recurrent neural networks and performs as a single task model.
- Single Task Learning GCN and GRU model (STL-GCN-GRU), which is built only upon the spatio-temporal blocks of our architecture.
- Multi Task Learning GRU (MTL-GRU)[18], the model represents the state-of-the-art for MTL traffic forecasting of flow and speed. The model is entirely based on GRU layers for the task-specific and shared representation. A merge layer combines information between tasks and allows to learn a deeper hidden representation.
- Spatiotemporal graph convolution model (TGCN)[21].
- Attribute-Augmented Spatiotemporal Graph Convolutional Network (AST-GCN)[24].
- Attention-based multi-task learning algorithm (AST-MTL)[1].

The hyperparameters of these baseline methods are the same as those in the original paper or released codes.

4.4 Experimental Results

4.4.1 Prediction Comparison with Baselines. In this experiment, we compare the performance of MT-C2G with the baseline methods to evaluate the prediction performance of our model. Aiming to check whether the algorithm will be biased towards the augmented data, we performed each experiences on all test dataset in addition to extreme weather class parts of the dataset only. The experimental results for the city of Calgary are shown in Tables 1 and 2. Due to the page limit, the RMSE curves of the MT-C2G on the Brussels data in comparison to other three deep learning methods is shown in Figure 5. The complete results for the city of Brussels are presented in Appendix A.

In general, MT-C2G obtains the best truck speed prediction performance under all evaluation metrics for all prediction horizons on both datasets. For example, for the 1 hour truck speed prediction horizon without performing oversampling, the RMSE of MT-C2G is reduced by approximately 8.6%, 1.8% and 2.6%, and the MAE error of MT-C2G is reduced by approximately 11.7%, 1.01% and 10.04%, in compare to the AST-MTL, AST-GCN and TGCN models respectively. Additionally, comparing MT-C2G performance with the STL-GCN-GRU which is built only upon the spatio-temporal blocks of the MT-C2G architecture, reveal the effect of considering correlation between truck speed and flow in the truck speed prediction task. As can be seen the RMSE and MAE of MT-C2G is reduced by approximately 1.76% and 6.02% in compare to STL-GCN-GRU model which proves that MT-C2G is able to perform the truck speed prediction better than baseline models. This results clearly show that performing multi-task learning and enhancing the model with contextual information, can facilitate truck speed prediction task.

Table 1: Performance comparison between MT-C2G and the baseline models for different prediction horizons for all weather classes - Calgary

Weather Condition	Over sampled	Model	1 hour		2 hours		3 hours		4 hours		5 hours		6 hours	
			RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
All Weather Classes	Without SMOTE	HA	11.613	8.714	12.053	9.053	12.374	9.305	12.601	9.480	12.775	9.601	12.927	9.705
		ARIMA	10.912	7.617	12.468	8.868	13.567	9.709	14.064	10.267	14.286	10.506	14.368	10.583
		SVR	10.461	7.079	11.498	7.987	12.118	8.495	12.514	8.846	12.698	8.950	12.833	9.015
		STL-GRU	10.833	7.713	11.633	8.233	12.075	8.560	12.298	8.725	12.485	8.872	12.677	9.066
		STL-GCN+GRU	10.037	7.044	10.182	7.050	10.219	7.124	10.327	7.240	10.273	7.161	10.111	7.088
		MTL-GRU	10.872	7.843	10.726	7.737	10.910	7.841	10.994	7.937	11.173	8.086	11.304	8.222
		T-GCN	10.132	7.3669	10.496	7.7636	10.9157	8.2613	11.240	8.278	11.442	8.463	11.610	8.543
		AST-GCN	10.040	6.690	10.349	7.583	10.394	7.446	10.674	7.755	10.730	7.832	10.752	7.815
		AST-MTL	10.792	7.508	11.485	7.995	11.346	7.893	11.650	8.143	11.785	8.338	11.937	8.364
		MT-C2G	9.8648	6.627	9.882	6.664	9.957	6.734	9.910	6.7649	10.153	6.986	10.314	7.118
	With SMOTE	HA	11.51	8.595	11.944	8.921	12.255	9.164	12.48	9.341	12.658	9.468	12.813	9.577
		ARIMA	10.940	7.665	12.463	8.870	13.549	9.721	14.060	10.316	14.280	10.543	14.378	10.612
		SVR	10.692	7.094	11.830	7.954	12.551	8.493	12.899	8.852	13.042	8.978	13.144	9.132
		STL-GRU	10.561	7.125	11.161	7.765	11.686	8.130	11.906	8.287	12.118	8.453	12.204	8.494
		STL-GCN+GRU	9.944	6.992	10.024	6.987	10.013	6.984	9.822	6.892	9.878	6.898	9.914	6.875
		MTL-GRU	10.344	7.311	10.826	7.532	11.335	7.886	11.548	8.038	11.754	8.199	11.837	8.239
		T-GCN	10.185	7.420	10.645	7.835	11.001	8.329	11.212	9.191	11.432	8.983	11.610	9.202
		AST-GCN	9.498	6.770	10.050	7.346	10.224	7.366	10.182	7.422	10.357	7.355	10.372	7.651
		AST-MTL	10.542	7.236	10.803	7.381	10.913	7.502	11.088	7.915	11.203	7.922	11.778	8.365
		MT-C2G	9.410	6.611	9.574	6.563	9.715	6.622	9.889	6.761	9.976	6.836	10.079	6.881

Table 2: Performance comparison between MT-C2G and the baseline models for different prediction horizons for extreme weather classes - Calgary

Weather Condition	Over sampled	Model	1 hour		2 hours		3 hours		4 hours		5 hours		6 hours	
			RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Extreme Weather Class	Without SMOTE	SVR	10.193	7.034	12.766	10.108	14.822	12.631	16.652	14.835	17.173	15.25	17.702	15.613
		STL-GRU	6.960	5.409	7.302	5.643	8.249	6.394	9.148	7.278	9.702	7.749	10.313	8.261
		STL-GCN+GRU	5.893	4.691	6.012	5.026	6.282	5.061	6.667	5.242	7.027	5.660	7.479	5.913
		MTL-GRU	6.827	5.211	7.155	5.417	8.001	5.946	8.69	6.667	9.410	7.516	10.04	7.76
		T-GCN	6.034	4.487	6.699	4.985	7.412	5.629	8.535	6.703	9.271	7.564	9.552	7.742
		AST-GCN	5.962	4.716	6.165	5.058	6.063	5.125	6.608	4.959	6.648	4.968	6.718	5.069
		AST-MTL	6.149	3.360	6.898	3.526	7.081	3.6721	7.396	3.564	7.894	3.690	8.123	3.698
		MT-C2G	5.182	3.9646	5.2319	3.8193	5.2495	3.889	5.3595	4.0396	5.4758	4.0717	5.7499	4.4324
	With SMOTE	SVR	8.722	5.442	10.26	6.554	11.363	7.444	11.966	8.076	12.374	8.463	12.87	9.022
		STL-GRU	5.805	4.334	5.974	4.444	6.27	4.65	6.694	4.988	6.698	5.094	6.77	5.166
		STL-GCN+GRU	5.394	4.001	5.416	4.078	5.5254	4.166	5.5701	4.194	5.945	4.541	5.925	4.650
		MTL-GRU	5.613	4.081	5.742	4.25	5.762	4.239	5.828	4.212	5.89	4.463	5.971	4.653
		T-GCN	5.491	3.819	5.532	3.926	5.366	3.962	5.608	4.324	5.68	4.434	6.019	4.78
		AST-GCN	5.384	3.759	5.395	3.81	5.35	3.952	5.439	3.919	5.476	4.181	5.528	4.289
		AST-MTL	5.945	3.789	6.106	3.868	6.267	3.957	6.451	4.121	6.864	4.307	7.054	4.534
		MT-C2G	4.8423	3.3607	4.9655	3.5263	5.0333	3.6721	5.0598	3.5644	5.2335	3.6904	5.2367	3.6988

Looking at the results for other time horizons, we observe that generally the performance of all models deteriorates with the increase of the prediction horizon. However, MT-C2G still outperforms other baselines under each prediction horizon. MT-C2G and

the AST-GCN show the best performance among all others which proves the impact of considering weather conditions during the truck speed prediction task. Additionally, as the prediction time

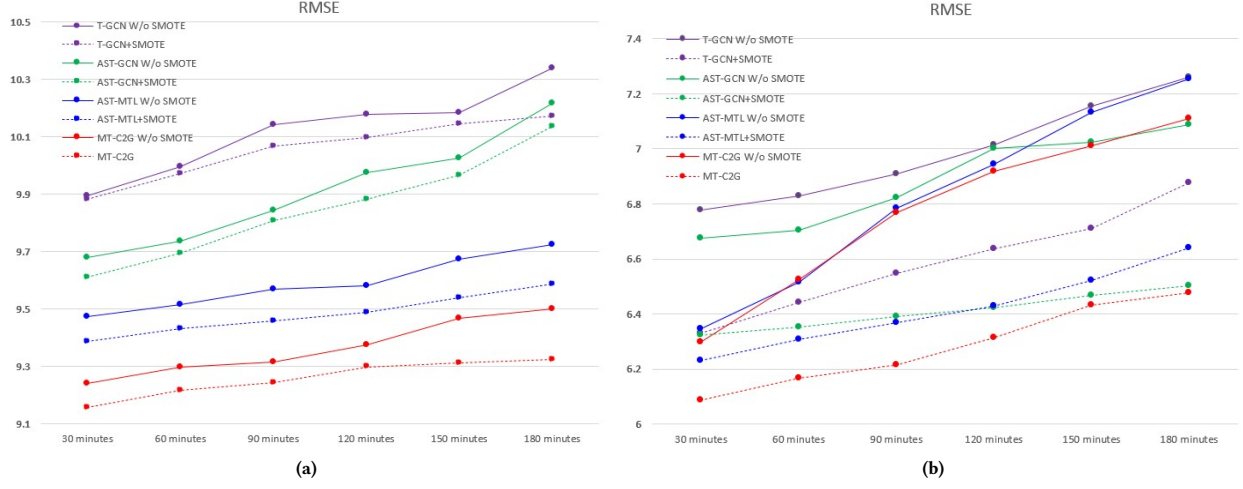


Figure 5: Performance comparison between MT-C2G and the baseline models for different prediction horizons - Brussels. (a) Changes in the RMSE on all weather classes (b) Changes in the RMSE for extreme weather class

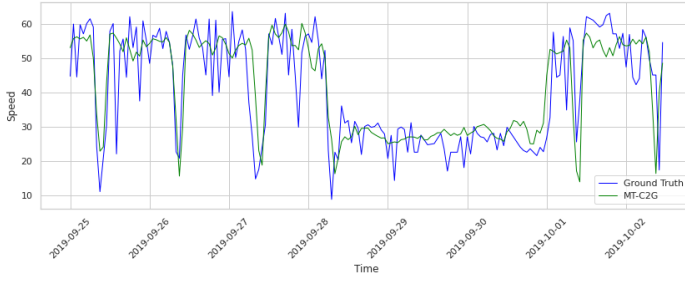


Figure 6: The visualization results for the 1 hour prediction horizon.

increases, the difference between MT-C2G and AST-GCN shows the performance of AST-GCN gradually becomes worse, and MT-C2G increases its improvement with the RMSE reduction from 1.8% to 4.07% and MAE reduction from 1.01% to 8.9%. This observation clearly shows the advantage of utilizing the MTL to capture the correlation between truck speed and truck flow in MT-C2G which makes it outperform other models in long-term prediction. Same pattern appears in the comparison of MT-C2G with AST-MTL and T-GCN. The improvement of MT-C2G starts with the RMSE reduction of 8.6% and 2.6% and ends with 13.59% and 11.16% in compare to AST-MTL and T-GCN respectively, and starts with an MAE reduction of 11.7% and 10.04% and ends with 14.9% and 16.68%. Based on the observations, we conclude that performing MTL and enhancing the model with contextual information, helps MT-C2G maintaining better performance for different prediction horizons compared to other baselines.

In order to gain a better understanding of the predictive performance of the MT-G2C model, we present a visualization of the ground truth and forecasting results for one road segment in the

city of Calgary dataset, as illustrated in Figure 6. The figure contains a comparison of truck speed predictions from MT-G2C and MT-G2C without SMOTE, for the 1 hour time horizon. It is evident that the MT-C2G is capable of tracking the changing trend of the real speed and, therefore, can accurately predict the truck speed. Furthermore, taking a look at the turning points and peaks, we can see that the MT-C2G did not perform well at certain points. One possible explanation for the sudden changes at peaks could be attributed to factors other than the contextual factors considered in this study, such as traffic accidents and road construction.

4.4.2 Prediction Comparison Under different Weather Conditions. The object of this experiment is to evaluate the performance of these models under the extreme weather conditions. To accomplish this goal we test all models on just extreme weather class. Table 2 shows the proposed model achieves the best performance compared to other models for traffic speed prediction under extreme weather conditions. For example, at the first time horizon and without oversampling, the RMSE of AST-GCN is 5.962, and for MT-C2G, it is 5.182, which shows a 13.08% reduction. Also, the MAE value decreased by 15.93% compared to AST-GCN for MT-C2G. Therefore, experimenting on just extreme weather class shows that MT-C2G captures the correlation between this class and truck speed more effectively. Also, Table 2 shows the average values of RMSE and MAE in extreme weather conditions for each forecast horizon. The proposed method outperforms all other methods under extreme weather conditions in RMSE and MAE. Furthermore, experiments shows the advantage of utilizing the GRU and MTL along with weather conditions as contextual factors, especially for truck speed prediction under extreme weather conditions.

4.4.3 Prediction Performance with and without Oversampling. SMOTE is used to make up for the under-represented truck data in extreme weather conditions. This experiment shows the comparison between oversampled and original data. An interesting result in the Table 1 and Table 2 is the improvement of prediction

performance after oversampling the data by the SMOTE method. As shown in Table 1, performing oversampling on the data helps almost all models (except SVR, ARIMA, and TGCN models in all weather classes experiment) to better capture the dependency between weather conditions and traffic status. For example, for 1 hour truck speed prediction horizon on all weather classes, after performing the oversampling by SMOTE, the RMSE and MAE of prediction decrease by 4.61% and 0.24% respectively in MT-C2G. The results for AST-MTL and AST-GCN models also proves that performing oversampling helps increasing the accuracy and precision of the models. Table 2 further highlights the benefit of using oversampling to balance the extreme weather classes before the prediction. In particular, balancing the the extreme weather class helps all models to perform better especially during the extreme weather conditions. As shown in the experiment, MT-C2G accurately captures the extreme weather correlation to truck speed. Therefore, using GRU and MTL along with using oversampling extreme weather conditions by SMOTE helps MT-C2G to have the best prediction performance.

5 CONCLUSION

In this paper, we propose a novel method called MT-C2G for the truck speed prediction under the extreme weather conditions. MT-C2G integrates the MTL, GCN and GRU mechanisms, which captures the spatio-temporal correlations between truck speed, flow, weather conditions, and other contextual information for the truck traffic prediction. To deal with imbalanced data with regards to extreme weather conditions, MT-C2G utilizes the SMOTE data augmentation method. The experiments are performed on two real world datasets, i.e. Calgary and Brussels, to evaluate the proposed MT-C2G. Experimental results show that the MT-C2G outperforms other state-of-the-art traffic prediction models. This study specifically highlights: (1) The truck speed prediction task can be improved by considering contextual factors together with the historical truck traffic data, (2) MT-C2G takes advantage of the commonalities between these two variables to boost the prediction accuracy through joint and simultaneous optimization of a shared model and using an attention mechanism that captures the commonalities, and (3) Solving challenge of under-represented data can significantly improve the performance of the model.

Truck traffic data is sparse and difficult to acquire. Other types of the traffic data, such as traffic count, traffic speed at intersections, is usually available through traffic monitoring systems. In addition, some non-recurrent events such as accidents, have great impact on the traffic, including truck speed. Therefore, it is worth investigating how to make use of and integrate other traffic data and non-recurrent events on the road networks in truck speed prediction. Furthermore, SMOTE is used in MT-C2G to deal with imbalanced weather data. One other future research direction is to explore and develop new data augmentation method that better suits spatio-temporally correlated truck traffic data.

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A DETAILED RESULTS FOR BRUSSELS DATA

Table 3: Performance comparison between MT-C2G and the baseline models for different prediction horizons for all weather classes - Brussels

Weather Condition	Over sampled	Model	30 minutes		60 minutes		90 minutes		120 minutes		150 minutes		180 minutes	
			RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
All Weather Classes	Without SMOTE	HA	10.872	7.084	11.04	7.210	11.207	7.344	11.371	7.484	11.533	7.629	11.696	7.784
		ARIMA	13.343	6.904	13.685	7.293	13.986	7.646	14.239	7.964	14.506	8.290	14.763	8.614
		SVR	11.242	5.682	11.352	5.788	11.451	5.885	11.543	5.973	11.634	6.070	11.723	6.165
		STL-GRU	10.423	6.714	10.545	6.792	10.667	6.832	10.729	6.896	10.870	6.957	11.005	6.971
		STL-GCN+GRU	9.818	6.188	9.832	6.265	9.923	6.389	9.932	6.375	10.029	6.497	9.960	6.423
		MTL-GRU	9.833	6.385	9.877	6.394	9.934	6.445	9.944	6.433	10.042	6.469	10.096	6.525
		T-GCN	9.895	6.125	9.996	6.235	10.143	6.326	10.179	6.387	10.184	6.399	10.341	6.693
		AST-GCN	9.679	5.917	9.736	6.079	9.845	6.077	9.976	6.204	10.026	6.327	10.217	6.494
		AST-MTL	9.473	6.260	9.515	6.269	9.570	6.319	9.580	6.307	9.674	6.342	9.726	6.397
		MT-C2G	9.242	5.925	9.298	5.977	9.315	5.979	9.375	6.040	9.467	6.059	9.500	6.117
	With SMOTE	HA	10.686	6.870	10.842	6.985	10.998	7.106	11.155	7.235	11.308	7.368	11.460	7.509
		ARIMA	13.362	6.963	13.676	7.319	13.965	7.648	14.232	7.972	14.515	8.308	14.780	8.641
		SVR	11.263	5.734	11.380	5.842	11.488	5.946	11.585	6.044	11.673	6.148	11.759	6.246
		STL-GRU	10.319	6.580	10.440	6.656	10.560	6.695	10.514	6.827	10.761	6.887	10.895	6.832
		STL-GCN+GRU	9.682	6.126	9.735	6.14	9.825	6.325	9.933	6.248	10.098	6.432	10.261	6.359
		MTL-GRU	9.735	6.257	9.778	6.266	9.735	6.381	9.845	6.304	9.841	6.404	9.894	6.46
		T-GCN	9.884	6.139	9.973	6.199	10.069	6.271	10.098	6.297	10.145	6.328	10.174	6.363
		AST-GCN	9.612	5.850	9.694	5.942	9.807	6.028	9.884	6.129	9.968	6.224	10.138	6.265
		AST-MT	9.386	6.015	9.433	6.016	9.458	6.029	9.488	6.078	9.540	6.142	9.587	6.200
		MT-C2G	9.156	5.803	9.216	5.848	9.245	5.902	9.299	5.938	9.313	5.947	9.325	6.025

Table 4: Performance comparison between MT-C2G and the baseline models for different prediction horizons for extreme weather classes - Brussel

Weather Condition	Over sampled	Model	30 minutes		60 minutes		90 minutes		120 minutes		150 minutes		180 minutes	
			RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
Extreme Weather Class	Without SMOTE	SVR	11.133	6.193	11.122	6.19	10.986	6.104	10.882	6.02	10.732	5.906	10.631	5.818
		STL-GRU	7.148	5.872	7.216	5.759	7.597	6.015	8.400	6.945	8.411	6.857	8.629	7.098
		STL-GCN+GRU	6.692	5.142	6.781	5.304	7.156	5.662	7.457	5.823	7.380	5.758	7.227	5.583
		MTL-GRU	6.934	5.637	6.855	5.586	7.293	5.774	7.98	6.598	8.159	6.651	8.37	6.743
		T-GCN	6.779	4.436	6.828	4.544	6.909	4.668	7.014	4.799	7.154	4.932	7.259	5.098
		AST-GCN	6.675	4.371	6.703	4.432	6.824	4.498	7.003	4.582	7.025	4.661	7.088	4.695
		AST-MTL	6.347	4.234	6.517	4.369	6.783	4.532	6.944	4.780	7.132	4.834	7.255	5.190
		MT-C2G	6.299	4.231	6.525	4.342	6.767	4.477	6.919	4.502	7.010	4.692	7.110	4.719
	With SMOTE	SVR	10.127	5.681	10.156	5.693	10.263	5.803	10.358	5.876	10.423	5.928	10.437	5.883
		STL-GRU	6.890	5.521	7.093	5.620	7.537	5.793	8.048	5.965	8.152	6.214	8.474	6.390
		STL-GCN+GRU	6.328	5.099	6.399	5.119	6.440	5.222	6.569	5.323	6.603	5.458	6.827	5.583
		MTL-GRU	6.746	5.310	6.838	5.395	7.236	5.561	7.446	5.686	7.626	5.765	7.751	5.944
		T-GCN	6.330	4.203	6.442	4.293	6.548	4.349	6.638	4.408	6.709	4.595	6.877	4.660
		AST-GCN	6.325	4.160	6.353	4.215	6.392	4.238	6.422	4.243	6.469	4.263	6.502	4.308
		AST-MTL	6.232	4.196	6.309	4.239	6.367	4.299	6.429	4.322	6.522	4.412	6.641	4.542
		MT-C2G	6.088	4.108	6.168	4.154	6.214	4.198	6.313	4.232	6.431	4.282	6.478	4.324